Q3: Mass Variation and a Cyclic Degeneration Obstruction

Research continuation based on the supplied October 8 manuscripts

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Abstract

The fixed objective is a smooth closed, non-Bach-flat global minimizer of the squared trace-free Ricci functional. We identify a useful freedom in one surviving degeneration: the marked smooth $\mathcal{O}(-3)$ filling admits scalar-flat Kähler ALE metrics of either mass sign at the same curvature integral. Combining this classical existence theory with the supplied critical-background boundary flux gives a first-order replacement formula. In a smooth critical orbifold limit, a marked $\frac{1}{3}(1,1)$ degeneration of this topology can remain minimizing only if a specific self-dual curvature eigenvalue vanishes. This is an obstruction to a degeneration, not a proof of smooth attainment.

1 The fixed existence problem

For an oriented smooth closed four-manifold, put

$$P(g) = \int_M |\operatorname{Ric}_g^\circ|^2 \, dV_g, \qquad p_*(M) = \inf_{g \in \operatorname{Met}(M)} P(g).$$

Here Weyl curvature has the Hilbert–Schmidt curvature-operator norm. The fixed candidate remains $M = X\#\overline{\mathbb{CP}^2}$, with X the classical Godeaux surface, $K_X^2 = 1$. The preceding work gives

$$2\chi(M) + 3\tau(M) = 0, \qquad \frac{16\pi^2}{3} \le p_*(M) < 8\pi^2. \tag{1}$$

The lower bound is the classical monopole estimate; the strict upper bound uses the supplied complete negative-Kähler–Einstein insertion construction. Equation (1) is inherited here.

All conformal classes in this example have negative Yamabe constant. A smooth P-minimizer has constant scalar curvature. If it were Bach-flat, its remaining Euler–Lagrange equation would give $s \operatorname{Ric}^\circ = 0$, hence it would be Einstein. But $T = 2\chi + 3\tau = 0$ and the Chern–Weil formula would then force $s = W^+ = 0$, contrary to the positive monopole scalar budget. Thus *smooth attainment*, without computation of $p_*(M)$, would answer Q3.

The preceding round found an actual embedded (-3)-sphere in X, disjoint from the chosen blowup. Its tubular neighbourhood is the disk bundle $\mathcal{O}(-3)$, with end $S^3/\mathbb{Z}_3$. Its usual scalar-flat Kähler ALE metric has $P = 8\pi^2/3$; simply comparing this number with (1) does not exclude a bubble. The present calculation tests *replacement at finite scale*, rather than adding independent lower bounds for the core and the bulk.

2 A fixed filling with arbitrary mass sign

Let N be the oriented smooth total space of $\mathcal{O}(-3)$, with generator $C \in H_2(N;\mathbb{Z})$, $C^2 = -3$, and scalar end group $\Gamma = \frac{1}{3}(1,1)$.

Proposition 1 (Consequence of classical ALE existence and mass theorems)**.** *There is a Stein complex structure J on this fixed marked smooth manifold such that, for every $t \in \mathbb{R}$, an ALE scalar-flat Kähler metric g_t exists with*

$$\int_C \omega_t = t, \qquad c_1(N,J)\cdot C = -1, \qquad m(g_t) = -\frac{t}{9\pi}, \qquad P(g_t) = \frac{8\pi^2}{3}. \tag{2}$$

Mass is normalized as in Hein–LeBrun [3].

Proof. In $\mathbb{F}_1$, write $\Gamma_0^2 = -1$, $f^2 = 0$, $\Gamma_0 f = 1$. Choose a smooth section $D \in |\Gamma_0 + 2f|$. It is ample and has $D^2 = 3$. Its complement is affine, hence Stein, and is an affine $\mathbb{C}$-bundle with underlying smooth bundle $\mathcal{O}(-3)$. Honda's Theorem 1.2 [1] realizes this complex structure in the ALE scalar-flat Kähler deformation family of the same smooth space. The compact generator has image $C = \Gamma_0 - f$ in $H_2(\mathbb{F}_1)$, so

$$C^2 = -3, \qquad c_1 \cdot C = (2\Gamma_0 + 3f)(\Gamma_0 - f) = -1.$$

Han–Viaclovsky's Theorem 1.13 [2] explicitly includes the group $\frac{1}{3}(1,1)$ and supplies a scalar-flat Kähler ALE metric in *every* Kähler class for each complex structure in this Artin family. Their Remark 6.1 identifies the ALE Kähler cone of this Stein surface with all of $H^2(N;\mathbb{R})$. Hence the period t is unrestricted. Positivity of a Kähler form does not require $t > 0$: C is not holomorphic in the Stein complex structure.

The intersection matrix is $Q = [-3]$. Hein–LeBrun's Theorem 4.3 gives

$$m = -\frac{1}{3\pi}(-1)Q^{-1}t = -\frac{t}{9\pi}.$$

The Ricci form ϱ is anti-self-dual and represents $2\pi c_1$. The relative square is $c_1^2 = -1/3$. Its ALE decay permits the relative Chern–Weil pairing, and therefore

$$P = 2\int_N |\varrho|^2 = -2\int_N \varrho \wedge \varrho = -8\pi^2 c_1^2 = \frac{8\pi^2}{3}.$$

The end marking is retained by the smooth bundle identification and the fixed ALE background used in the all-class existence theorem. □

This proposition imports a concrete freedom from established theory; it is not a claim that mass-sign flexibility is historically new. Its relevance is that a replacement metric need not stay within the radial negative-mass LeBrun family.

3 The leading tail and its normalization

Let J_0 be the standard complex structure on the end, and set

$$B(x) = \frac{|x|^2 I - xx^{\mathsf{T}} - (J_0 x)(J_0 x)^{\mathsf{T}}}{|x|^4}. \tag{3}$$

Han–Viaclovsky, Remark 2.2, supplies differentiated decay $g - \delta = O(r^{-2})$, $J - J_0 = O(r^{-3})$. Yao's Theorem D(ii) [4] then gives a logarithmic Kähler potential whose coefficient is determined by mass, with $O(r^{-3})$ metric remainder. In suitable end coordinates,

$$g_t = \delta + 9m(g_t)B + q_t, \qquad |\partial^j q_t| \le C_j r^{-3-j}. \tag{4}$$

This assertion concerns a fixed chosen g_t; no uniform estimates over all t are required.

The coefficient in (4) is also checked directly. For $h = aB$,

$$\partial_j h_{ij} - \partial_i \operatorname{tr} h = \frac{2ax_i}{r^4}.$$

The ADM flux with coefficient $1/(12\pi^2)$ on $S^3/\mathbb{Z}_n$ is $a/(3n)$. For $n = 3$, $a = 9m$. Yao's cone-link convention uses $m_Y = 3m$ for this end; (4) avoids any ambiguity from this different convention or from d^c.

The $O(r^{-3})$ terms may contribute at later orders, but there is no independent leading r^{-2} tensor whose sign needs to be guessed. This is why the mass formula can be used in the full metric replacement, not merely in a radial model.

4 Replacement of a smooth critical orbifold point

Write

$$R^+ = W^+ + \frac{s}{12}I, \qquad \mathcal{C}(g) = \int \left(4|W^+|^2 + \frac{s^2}{12}\right) dV = 4\int |R^+|^2 dV.$$

On a closed manifold, $P = \mathcal{C} - 8\pi^2 T$, $T = 2\chi + 3\tau$; on pieces the corresponding boundary terms must be retained.

Proposition 2 (First-order replacement; inherited flux normalization)**.** *Let g be a smooth orbifold metric near a point of type $\frac{1}{3}(1,1)$, and suppose it is P-critical on a neighbourhood containing the full support of the modification. Let ζ be the unit self-dual form singled out by the isotropy representation, and put*

$$\mu = \langle R^+\zeta, \zeta\rangle_p.$$

Replace the point by the marked filling (N, g_t) at length scale λ. There are smooth matched competitors, equal to g outside that neighbourhood, for which

$$P(g_{\lambda,t}) = P(g) + \frac{8\pi^2}{3} - 24\pi^2 m(g_t)\mu\lambda^2 + o(\lambda^2) = P(g) + \frac{8\pi^2}{3} + \frac{8\pi}{3}t\mu\lambda^2 + o(\lambda^2). \tag{5}$$

Here the linear boundary flux is the corrected critical-background flux from supplied source A02.59D. The following proof verifies the transfer to the nonradial filling (4) and the error order.

Proof. In orbifold normal coordinates, $g = \delta + H_2 + O(r^3)$, with H_2 the quadratic curvature jet. The scaled ALE end is

$$g_{t,\lambda} = \delta + 9m\lambda^2 B + q_{t,\lambda}, \qquad |\partial^j q_{t,\lambda}| \le C_j \lambda^3 r^{-3-j}.$$

Choose $\rho = \lambda^{3/7}$. On the inner piece use the exact scalar-flat Kähler metric, add the background jet, and extend that jet smoothly over the core. In unscaled core coordinates this extension is $O(\lambda^2)$ with bounded derivatives. Its physical chiral curvature is $O(1)$ on a region of volume $O(\lambda^4)$. On the end, relative to the common background chiral curvature, the errors are $O(\lambda^2 r^{-2}) + O(\lambda^3 r^{-3}) + O(r)$.

After subtracting the removed bulk integral, the inner error is bounded by

$$C\left(\lambda^4 + \lambda^2\rho^2 + \lambda^3\rho + \lambda^4|\log\lambda| + \rho^5\right). \tag{6}$$

This subtraction is essential: the common $O(\rho^4)$ background term is present in both integrals. Metric norms and volume factors contribute to the displayed bounds as well.

In the outer region, expand about the exact bulk metric. Its interior first variation vanishes by criticality. The mass term therefore gives only the inner boundary flux. The supplied value $-8\pi^2\mu$ for a unit Burns tail is linear in its coefficient, and quotient integration divides by three. Thus the leading boundary term is

$$-\frac{8\pi^2}{3}(9m)\mu\lambda^2.$$

Changing the curvature on the inner sphere to its value at p costs $O(\lambda^2\rho)$. The outer quadratic remainder is $O(\lambda^4\rho^{-4})$. The differentiated remainder in (4) has linear flux $O(\lambda^3/\rho)$, squared

error $O(\lambda^6\rho^{-6})$, and mixed error bounded by $O(\lambda^5\rho^{-5})$. All are $o(\lambda^2)$, as are (6): for example, $\rho^5 = \lambda^{15/7}$ and $\lambda^4\rho^{-4} = \lambda^{16/7}$.

Finally, the relative T contribution of the filling is $-1/3$. Consequently the fixed difference between the P and chiral-functional comparisons is $8\pi^2/3$. This proves (5). The argument uses a fixed mass and a fixed filling before $\lambda \downarrow 0$; it does not exchange these limits or assume an Einstein bulk. □

5 The necessary condition for a minimizing degeneration

Corollary 3. *Suppose an ideal minimizing configuration has a smooth critical orbifold bulk with a $\frac{1}{3}(1,1)$ point, energy equality, and a reconstruction on the original manifold using a marked $\mathcal{O}(-3)$ filling. Then*

$$\boxed{\mu = 0.} \tag{7}$$

Proof. The relative identity gives $P(N) \geq 8\pi^2/3$ for any admissible filling with this topology and asymptotics, since its chiral energy is nonnegative. Replacing it by (2) shows that an ideal minimizer must use exactly this value. If $\mu > 0$, use a positive-mass filling; if $\mu < 0$, use a negative-mass filling. For fixed such mass, (5) is strictly below the ideal energy for all sufficiently small positive λ, contradicting minimality on the same smooth manifold. Other resolved pieces can be recovered afterwards with error smaller than this fixed improvement. □

The same comparison is local. A full global energy identity can be replaced by the following precise local hypotheses: the excised regions in the minimizing sequence have the marked $\mathcal{O}(-3)$ topology, their boundary data converge smoothly to the orbifold-ball boundary, and

$$\liminf_j P(U_j) \geq P_g(B_r/\Gamma) + 8\pi^2/3. \tag{8}$$

For any fixed small λ, (5) supplies a filling matching g near that boundary with a fixed strict saving. Smooth boundary convergence permits matching to g_j with error tending to zero. Leaving the exterior unchanged contradicts minimizing energy. Relative Chern–Weil and lower semicontinuity of the nonnegative chiral density yield (8) when the stated marking and boundary convergence are available. This argument does not require compactness of the exterior.

6 What is still required for Q3

For scalar $\mathbb{Z}_3$ isotropy, the self-dual block at the fixed point is

$$R^+ = \operatorname{diag}(\mu, \nu, \nu), \qquad \mu + 2\nu = s/4.$$

Equation (7) reduces the surviving block to

$$R^+ = \operatorname{diag}(0, s/8, s/8). \tag{9}$$

For $s < 0$ this is a legitimate algebraic curvature block. Negative scalar curvature alone does not exclude it. In the radial LeBrun calculation, the attained second-order logarithmic coefficient is proportional to $-\mu(\mu - s/8)$, so it vanishes in (9). That calculation must not be assigned without proof to all the nonradial fillings of Section 2. Their later-order interaction remains to be determined.

The result also does not identify every bubble with $\mathcal{O}(-3)$, establish smooth orbifold extension for every selected limit, or exclude Ricci-flat ALE and open or partially collapsed limits. Those are still part of the attainment problem. No smooth closed global minimizer has been constructed in this continuation.

The proved input from the literature is (2)–(4). The new comparison developed here is (5), using the specified inherited critical flux; its consequence is (7) under the stated limit and reconstruction hypotheses. Its concrete use is to replace the former one-sided restriction $\mu \geq 0$ by the equality $\mu = 0$, without altering the core energy or the marked smooth filling.

References

[1] N. Honda, *Deformation of LeBrun's ALE metrics with negative mass*, Theorem 1.2.
https://arxiv.org/abs/1204.4857

[2] J. Han and J. A. Viaclovsky, *Existence and compactness theory for ALE scalar-flat Kähler surfaces*, Forum Math. Sigma 8 (2020), e1. Theorem 1.13; Remarks 2.2 and 6.1.
https://arxiv.org/abs/1901.05611

[3] H.-J. Hein and C. LeBrun, *Mass in Kähler geometry*, Comm. Math. Phys. 347 (2016), 183–221. Theorem 4.3.
https://arxiv.org/abs/1507.08885

[4] Q. Yao, *Mass and Expansion of Asymptotically Conical Kähler Metrics*, Theorem D(ii).
https://arxiv.org/abs/2206.06505

[5] C. LeBrun, *Curvature Functionals, Optimal Metrics, and the Differential Topology of 4-Manifolds*, Section 11.
https://arxiv.org/abs/math/0404251

[6] Supplied research manuscripts: `01_Research_Synthesis.tex` and `02_Mathematical_Source_Companion.tex`, 8 October 2026. In particular, A02.59D for the corrected critical-background flux; the preceding continuation supplies its general critical-background form. These are project inputs, distinct from the published sources above.